

Our DNS infrastructure is now running. You can run the following lines:

```
$ dig SRV _ldap._tcp.network.local
$ dig SRV _kerberos._tcp.network.local
```

And both should return an answer section. If they do, congratulations—this DNS set-up works.

Final touches

Start by testing Kerberos functionality. Type in:

```
$ kinit administrator@NETWORK.LOCAL
```

The realm needs to be typed in upper-case. If it works, it should tell you something about the ticket expiring after a particular number of days (I was informed that mine would expire in 41 days). Also, if you type in `klist -e`, it will list the Kerberos ticket you've received.

We also installed an NTP server, because Kerberos is very fussy about time, and you can use the Samba box as a time server to let your clients sync their clocks with the AD server. It should be configured to use the NTP Pool Servers, so it doesn't need any configuration.

We're now ready to join our first computer to the AD.

Joining the domain

When you're joining the domain from a Windows box, type the domain name in upper-case (like we did with the `kinit` command). When asked for a username and password, type in 'administrator' as the username, along with the administrator password. Reboot the computer and log in to the domain by typing 'NETWORK\administrator' (replace NETWORK with your own WINS domain and make sure it's all upper-case) as the username, and the administrator password.

Administering the server

Samba 4 doesn't come with any administration tools, so how do you add, remove or modify users, computers, OUs, forests, etc? Well, you simply use Windows Server's official tools. Load up a client with the remote admin tools for Windows Server. If you're using Windows XP, install the Windows Server 2003 Service Pack 2 Administration Tools Pack, available at <http://bit.ly/PvjvSG>.

If you're using Windows Vista (you must run at least Service Pack 1), install the RSAT (Remote Server Administration Tools) for Windows Vista, available at <http://bit.ly/LtEdgk>.

If you're using Windows 7, install the RSAT for Windows 7 SP1 (you don't need SP1 to actually install the RSAT) from <http://bit.ly/NcnwmR>.

Be warned that while the 2003 and Vista tools weigh in at around 15-30 MB, the Windows 7 RSAT is a whopping 240 MB download *per architecture*.

With Windows 7, just installing the RSAT won't do—you

have to activate it by heading to Control Panel, searching for 'Turn On Windows Features', opening the applet and activating the entire 'Remote Server Administration Tools For Windows 7' group, and then rebooting the PC. It's almost as if using Windows 7 is a crime.

Once the remote admin tools are installed, you can use the classic `DSAMSC` to administer the Samba 4 server just as if it were a Windows Server. You can create forests, add backup servers to the forests (which may be more Samba servers, or Windows Servers—our set-up works with Windows Server 2008 R2 and should work just dandy with Windows Server 2012, when it comes out), and just generally do anything you wish without limitations—it's fully featured.

Samba 4 also supports Group Policy out-of-the-box, so you can launch `GPEDIT.MSC` and edit group policy to your heart's content.

Roaming profiles with Windows 7

If you want to support Windows 7 Roaming Profiles, you need to create the 'profiles' share and do some tinkering with `DSA.MSC`. First, open up `/etc/samba/smb.conf` (on the Samba box) and include the following lines:

```
[profiles]
    path = /var/samba/profiles
    read only = no
```

Create the profile directory, as follows:

```
$ sudo mkdir /var/samba/profiles
```

Then, start up `DSA.MSC`, select all the users you want to add roaming profile support to, right-click the selection and click *Properties*. In the *Profile* tab, locate the *Profile Path* input field, and type:

```
\\server.network.local\profiles\%USERNAME%
```

Replace `server.network.local` with your Samba box's DNS name. And that's it.

So what's next?

Well, play around with it. Remember, you can also add Macs to an AD set-up. Mac OS X has built-in support for AD. Linux computers can also join AD—you can use either Centrify or Likewise Open to do the job.

That's all from me. Adios! 

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